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Definitions, measures, and uses of rurality: A systematic review of the empirical and quantitative literature

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ABSTRACT

Definitions and measures of rurality are important to researchers and policymakers as they have implications for policy and practice. By conducting a systematic review, we provide a comprehensive overview of the various ways in which rurality has been defined and operationalized in empirical and quantitative analyses. We describe how descriptive definitions and measures of rurality have changed over time, particularly due to global and regional urbanization and advances in statistical techniques and data availability. We also provide a comprehensive breakdown of the dimensions and components of rurality measures, an overview of the techniques used in constructing rurality measures, and discuss the uses of rurality measures in research and policy. Finally, we discuss current limitations of descriptive rurality research, future directions, and implications for policy and practice.

1. Introduction

Definitions and measures of rurality are important to policymakers, researchers, and communities as they have strong implications for policy and practices across governing levels. Issues of community sustainability, well-being, equity, and access to services and amenities are intimately tied to rural life (Brown and Cromartie, 2004; Minore et al., 2008), and as modern lives and expectations shift, these issues are only becoming more salient to rural communities. For instance, over the past several decades, worldwide rural populations have steadily declined due to an increase in migration to urban locations resulting from limited job opportunities and modern conveniences in some rural communities (Bell and Osti, 2010). From 1950 to 2018, the global urban population has shown an increase of approximately 3.5 billion (UN DESA, 2018). By the year 2050, the urban population is expected to increase another 13 percent, rising from 55 percent to 68 percent (UN DESA, 2018).

These gradual, but nevertheless drastic, shifts in population distribution can have profound consequences within vital areas such as healthcare resource allocation, political influence, educational outcomes, economic development, and agriculture production. For instance, in education, the primary concern is access to schooling, staffing of schools, school attendance, and financial support of schools (Crouch and Nguyen, 2020; Beeson and Strange, 2003; Sher, 2019). Students in rural areas are not only less likely to attend a postsecondary institution but are also less likely to complete their education to obtain a degree (Maltzan, 2006). In economic development, the concerns are primarily associated with job growth and retention, job diversification, industry attraction, labor force availability, labor force skills, tax base and incomes (Ellis and Biggs, 2001; Galluzzo, 2018). In the healthcare arena, concerns are associated with well-being of people in rural communities, qualifications and quantity of healthcare workers, access to healthcare facilities, and insurance coverage (Higgs, 1999; Inagami et al., 2016; Kulig et al., 2008; Mao et al., 2015; Zhao et al., 2019). From an environmental sustainability standpoint, rurality is associated with issues of land and water resource exploitation (Brown et al., 2005; Chen et al., 2018). Many of these issues are not solely confined to rural areas, however it has long been argued that rural areas tend to display or experience these issues for different reasons than say inner-city areas (Berenguer et al., 2005; Ellis and Biggs, 2001). For example, teacher turnover is a large concern nationally, however teacher attrition issues are more elevated in rural and geographically isolated states (Crouch and Nguyen, 2020).

Recognizing that rural areas face unique challenges, rural-specific development and policy initiatives, such as the United States

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Department of Agriculture Rural Development agency, which provides financial support for rural community development projects (Vilsack, 2016), have been promoted in many countries. Implementation of rural-specific policies and practices often relies on quantitative measures designating where policies and practices are applicable often implicitly consider rural areas to be homogenous and subject to the same conditions (National Academies of Sciences, Engineering, and Medicine, 2016). However, even within a single country, rural areas are heterogeneous. For example, while outmigration from rural communities was common across much of the agricultural-producing regions of the United States between 2000 and 2010, during this same time population growth was common in rural areas seen as having more outdoor recreation opportunities (Johnson, 2017). Therefore, how rurality is defined and measured matters greatly for policy and practice and consequently for the people who live there.

To develop effective planning and policies that address the well-being and sustainability of rural communities, tangible definitions of rurality are needed. Rurality has been defined in many ways, most of which can be organized into either descriptive definitions or socio-cultural definitions. Since our emphasis is on how rurality has been used for policy and practice, we focus our attention on the descriptive definitions of rurality, or ways of defining rurality based on what is observable and measurable, instead of socio-cultural definitions that are based on people's socio-cultural characteristics, their perceptions and construction of rurality, and the type of social environment in which they live (Halfacree, 1993; National Academies of Sciences, Engineering, and Medicine, 2016; Woods, 2009). While descriptive definitions of rurality have been strongly critiqued by rural scholars as providing a very limited view of the social construction that is rurality, and in many cases simply being a reflection of what the researcher expects to find (Cloke, 1994; National Academies of Sciences, Engineering, and Medicine, 2016; Woods, 2009), quantitative measures remain popular with both researchers and policy makers as they provide opportunities to assess and develop practical applications at scale.

In the last several decades, the most commonly used rurality measures focus on population density and/or distance from metropolitan areas (Minore et al., 2008; Mountrakis et al., 2005). For example, the United States Census Bureau and the United States Department of Agriculture have used population thresholds, such as having at least 50,000 individuals, to determine what is considered urban versus rural (Cromartie and Bucholtz, 2008). With such broad classification, there are issues of accurate policy implementation and resource allocation for areas that receive a rural/non-rural classification. For instance, it has long been recognized that there are substantial disparities in health outcomes between urban and rural areas, attributable to a variety of factors such as access to health services, basic amenities and residential mobility (Doogan et al., 2018; Zhao et al., 2019). It is possible to address these inequities, but more nuanced definitions of rurality are needed as scholars have long argued that ill-defined distinctions between rural and urban spaces lead to ill-defined and poorly-implemented policies (Isserman, 2005; Partridge et al., 2009; Waldorf, 2006). This is increasingly important given that in recent years, many of the distinctions between rural and urban areas have blurred (Jacquet et al., 2017; Lichter and Brown, 2011; National Academies of Sciences, Engineering, and Medicine, 2016). Targeted policies and precise implementation are more likely to be successful than those that are overly broad, particularly given limited resources (Dubois, 2012; Howlett, 2009; Partridge et al., 2009).

To address these challenges and shortcomings, many quantitative definitions and methods of defining rurality, such as the construction of indices containing multiple characteristics of rurality, have been proposed and developed. Given that situational contexts of the rural vary across the globe, as well as within individual countries, and that different rurality definitions have different purposes, it is widely acknowledged that there is no single index, set of factors, or scale of analysis that is universally appropriate for describing rurality in all locations and for all purposes (Doogan et al., 2018; Isserman, 2005; National Academies of Sciences, Engineering, and Medicine, 2016). However, the lack of consistency in rurality measures hampers comparisons of studies of rurality as well as rural policy analysis and evaluation (National Academies of Sciences, Engineering, and Medicine, 2016). Moreover, while there is reason to support the development of unique measures based on localized contexts, frequently the selection of specific variables for rurality measures is not explicitly grounded in a guiding theory or conceptualization of the rural, and may perhaps be more a reflection of conveniently available data than an actual representation of the nature of rurality for a specific location.

Given the diversity of quantitative rurality measures, there is a need to determine what characteristics or components of rurality, units of measurement (scale), and operationalization methods are most consistently used in order to identify the strengths and shortcomings of existing quantitative conceptualizations and operationalizations of rurality. Identification of these strengths and shortcomings will allow for the advancement of the research on measures of rurality and their application to policy and practice. To address these gaps, we conduct a systematic review of the descriptive definitions and empirical measures of rurality and how these quantitative conceptualizations and operationalizations have been used in policy and practice.

The rest of the paper is organized as follows. In the next section, we discuss our systematic search process, which includes the eligibility criteria, the literature search, and the coding of the reports. Then we briefly provide a historical summary of how rurality has been defined and measured descriptively. We focus the discussion on how advances in geospatial processing and data availability have fundamentally changed how rurality is conceptualized and operationalized quantitatively, the various metrics and techniques used to create rural indices, how these different measures of rurality have been used, and the limitations and shortcomings of the research in this area. We conclude with a discussion on directions for future research and the policy and practical implications of this research for policymakers and scholars.[2]

2. Methods

Our systematic review is designed to examine the conceptualization and operationalization of descriptive definitions and measures of rurality. To define the eligibility criteria, literature search, coding, and reporting conventions, we follow the Preferred Reporting Items for Systematic Reviews and Meta-Analysis standards as defined by Moher et al. (2009). We refer to studies included in our systematic review as *primary studies*.

2.1. Eligibility criteria

Our study endeavors to include all searchable studies that have explored rurality as a descriptive definition to provide a thorough and systematic examination of how rurality has been defined and measured quantitatively and empirically. Thus, the primary studies eligible for inclusion in this systematic review need to meet the following criteria: (a) the study defines or conceptualizes rurality as more than just the binary counterpoint to urban-suburban areas; or (b) the study proposes

[2] The rurality literature is admittedly much more vast and complex than that discussed here, however an in depth description of the rurality literature is outside the scope of this study. We refer readers to Halfacre (2001), Woods (2009), Kūle (2008), Racher et al. (2012), and Koziol et al. (2015), to name a few, for in-depth discussion of the literature, particularly around subjective and constructivist perspectives of rurality. Our aim instead is to provide a systematic examination of descriptive rurality measures and definitions, focusing on providing detailed quantitative and descriptive summary information on rurality measures to inform development of new rurality measures and identify trends and gaps within the empirical literature.

or develops a new measure of rurality or modifies an existing measure of rurality, including a description of the characteristics or components of said rurality measure.

2.2. Literature search

We obtained primary studies from searching commonly used economic and general social science databases, including ERIC, JSTOR, DOAJ, WorldCat, ProQuest, Google Scholar, and Taylor and Francis online. Through an iterative process, we created the following search string: ("index of rurality" OR "rurality index" OR "relative rurality" OR "rurality measure" OR "define rural" OR "conceptualize rurality" OR "measuring rurality"), which returned nearly 5000 studies. We also searched for "grey" literature using Dissertation and Thesis Repositories in WorldCat and ProQuest. We did not limit our search publication date or location, but we limited our search to full-text studies available in English. Our official search ended November 2019.

We note that, due to the nature of the search and our own limitations, particularly that we only know English well enough to do a systematic search of English language studies and most search engines have language limitations, our search will have more results for Anglophone contexts and underrepresent other contexts. This, in particular, can reduce the variations in how identified studies conceptualize and examine rurality (Gkartzios et al., 2020), which is an inherent drawback to most systematic reviews and to ours in particular. In other words, even though our search is systematic, the primary studies from our search may still be biased in terms of variations and uses of rurality due to the limitations of systematic reviews in any specific language and the specific utilization of the terms "rural" and "rurality" in the construction and conceptualization of rural place and space. While this does not necessarily detract from the importance and applications of our findings, it is critical to note this limits the scope of our work and the extent to which our results can be generalized.

2.3. Studies meeting eligibility criteria

From the initial search of databases and previous reviews, we used a three-phase process to screen primary studies that met all eligibility criteria (Figure A1). In pairs, coders independently read the title, abstract, and introduction for all of the studies obtained in our original search using the search string. Each pair checked if there were studies that were retained by one coder but not the other and discussed their reasons for including or excluding studies. The first author then checked over each pair's work and resolved any discrepancies or disagreement. We retained a study if the title, abstract, or introduction mentioned that the study defined rurality, or proposed or created new measures of rurality. We examined nearly 5000 records to ensure we captured the relevant studies on rurality (Table A1). The majority of these studies, however, do not define rurality or propose or create new measures of rurality.

In phase two, we were left with 215 studies for full text reading. Using Rayaan (Ouzzani et al., 2016) pairs of coders independently assessed whether each study fit the eligibility criteria outlined above. The coders discussed any discrepancies and made exclusion decisions together. The first author resolved any disagreement between coders. From these fully reviewed studies, we excluded studies that did not define or propose or create new measures of rurality. For example, Koziol et al. (2015) provide an extensive overview of operational definitions of rurality and demonstrate application of three existing rurality measures, discussing the differences in findings across measures, however do not propose a novel definition or measure of rurality. Similarly,

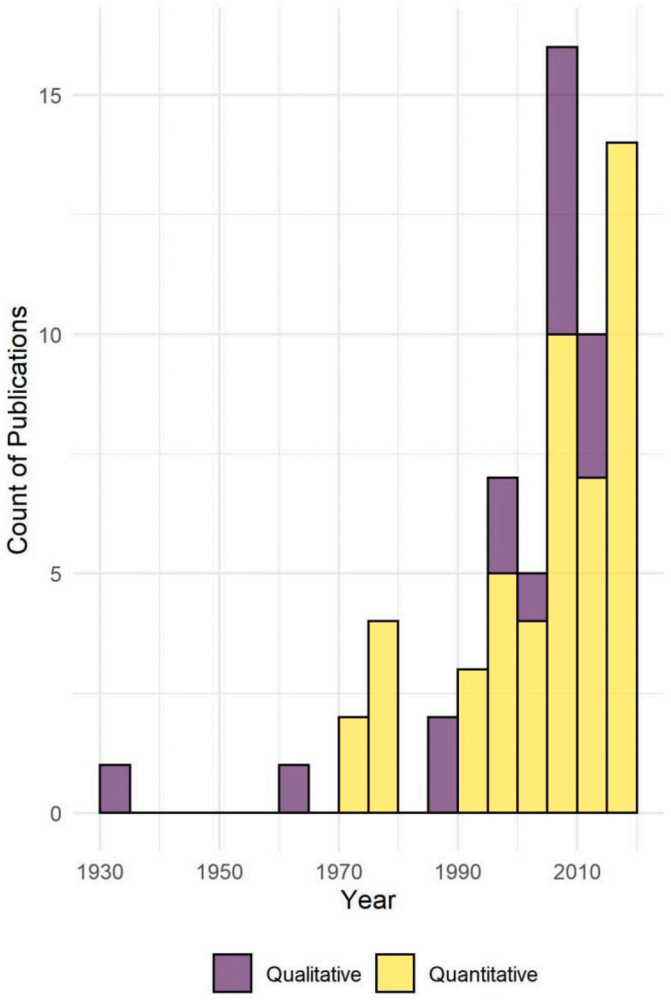


Fig. 1. Publications that define rurality by quantitative v. qualitative methods over time (binned into 5-year intervals).

Waldorf and Kim (2015) compare multiple rurality measures but do not propose a new measure. We also excluded sources for which no digital full-text in English was available such as *Ruralidad de la Región Centro-Este de México* (Zamudio Sánchez et al., 2009) and *Measuring rurality: the case of Texas* (Carney et al., 1990), for which rurality measure information available was incomplete or unclear. After screening, we were left with a sample of 65 studies that met all eligibility criteria.

2.4. Coding reports

The first author trained pairs of coders on how and what to code using two studies as examples. Then pairs of coders independently coded relevant information for each of the eligible studies using a common coding schema. Some of the key information include how rurality was defined or conceptualized, the measure/operationalization and components of rurality, quantitative techniques and formulae used to create the index (if applicable), geographic location(s), geographic unit of measurement, data source and year, policy implications, and main findings from the study (Table A2). Disagreements were resolved within pairs by consensus. As an additional check, the first author also examined the coding for consistency and accuracy across all studies.

Table 1
Component table for studies with a quantitative rurality index or typology.

Component/Dimension	Studies	Count of Studies	Percent of Studies
Population density	Berry et al. (2000); Beynon et al. (2016); Blunden et al. (1998); Bogdanov et al. (2008); Caschili et al. (2015); Cleland (1995); Cloke (1977); Cohen et al. (2017); Cromartie and Swanson (1996); Dická et al. (2019); Doogan et al. (2018); Fan and Wang (2018); Gajić et al. (2018); Harrington & O'Donoghu (1998); Inagami et al. (2016); Isserman (2005); Kralj (2009); Lauwers et al. (2005); Li et al. (2015); Madu (2010); Mao et al. (2015); Mao et al. (2018); Mitchell and Doyl (1996); Ocaña-Riola and Sánchez-Cantalejo (2005); Öğdül (2010); Openshaw and Coombes (1991); Pászto et al. (2015); Prieto-Lara and Ocaña-Riola (2010); Romano et al. (2016); Shoultz et al. (2007); Smith and Parvi (1973); Smith and Parvin (1975); Waldorf (2006); Zhao et al. (2019)	34	69%
Population	Berry et al. (2000); Caschili et al. (2015); Cohen et al. (2017); Cromartie and Swanson (1996); Cromartie et al. (2013); Dická et al. (2019); Inagami et al. (2016); Kralj (2009); Leduc (1997); Nong (2015); Nutley (1980); Öğdül (2010); Olatunde et al. (2007); Openshaw and Coombes (1991); Pászto et al. (2015); Smith and Parvin (1973); Terashima et al. (2014); Waldorf (2006); Weinert and Boik (1995)	19	39%
Population Percent in Urban or Rural Areas	Berry et al. (2000); Cohen et al. (2017); Cromartie and Swanson (1996); Isserman (2005); Li et al. (2015); Madu (2010); Peng et al. (2016); Riddick and Leadley (1978); Smith and Parvin (1973); Smith and Parvin (1975); Waldorf (2006)	11	22%
Population Change	Berry et al. (2000); Beynon et al. (2016); Blunden et al. (1998); Bogdanov et al. (2008); Caschili et al. (2015); Cleland (1995); Cloke (1977); Cromartie and Swanson (1996); Dická et al. (2019); Gajić et al. (2018); Hedlund (2016); Li et al. (2015); Pászto et al. (2015); Romano et al. (2016); Smith and Parvin (1973)	15	31%
Distance to Urban Area	Blunden et al. (1998); Cleland (1995); Cohen et al. (2017); Cromartie and Swanson (1996); Cromartie et al. (2013); Dická et al. (2019); Harrington & O'Donoghue (1998); Hoffmann et al. (2017); Inagami et al. (2016); Jacquet et al. (2017); Mitchell and Doyle (1996); Nielsen and Johansen (2009); Openshaw and Coombes (1991); Pászto et al. (2015); Smith and Parvin (1975); Waldorf (2006)	16	33%
Developed Land Cover	Blunden et al. (1998); Caschili et al. (2015); Cloke (1977); Cromartie and Swanson (1996); Hoffmann et al. (2017); Mao et al. (2015); Mao et al. (2018); Nielsen and Johansen (2009); Nong (2015); Öğdül (2010); Pászto et al. (2015); Romano et al. (2016); Shoultz et al. (2007); Terashima et al. (2014)	14	29%
Agricultural, Forested or Natural Land	Berry et al. (2000); Bogdanov et al. (2008); Gajić et al. (2018); Lauwers et al. (2005); Madu (2010); Nong (2015); Peng et al. (2016); Romano et al. (2016); Zhao et al. (2019)	9	18%
Topography	Blunden et al. (1998); Bogdanov et al. (2008); Gajić et al. (2018); Nutley (1980); Romano et al. (2016)	5	10%
Infrastructure and Service Accessibility or Density	Blunden et al. (1998); Bogdanov et al. (2008); Caschili et al. (2015); Cleland (1995); Cloke (1977); Cromartie and Swanson (1996); Dická et al. (2019); Doogan et al. (2018); Fang and Wang (2018); Gajić et al. (2018); Harrington & O'Donoghue (1998); Kralj (2009); Leduc (1997); Madu (2010); Mao et al. (2015); Mao et al. (2018); Mountrakis et al. (2005); Nong (2015); Nutley (1980); Öğdül (2010); Olatunde et al. (2007); Romano et al. (2016); Terashima et al. (2014); Zhao et al. (2019)	24	49%
Age Demographics	Bogdanov et al. (2008); Cleland (1995); Cloke (1977); Dická et al. (2019); Harrington & O'Donoghue (1998); Hedlund (2016); Li et al. (2015); Mitchell and Doyle (1996); Ocaña-Riola and Sánchez-Cantalejo (2005); Prieto-Lara and Ocaña-Riola (2010); Romano et al. (2016)	11	22%
Sex Demographics	Beynon et al. (2016); Bogdanov et al. (2008); Harrington & O'Donoghue (1998); Hedlund (2016); Öğdül (2010)	5	10%
Race/Ethnicity, Religion or Immigrant Demographics	Gajić et al. (2018); Mao et al. (2015); Mao et al. (2018)	3	6%
Agricultural Production	Blunden et al. (1998); Bogdanov et al. (2008); Caschili et al. (2015); Cloke (1977); Gajić et al. (2018); Harrington & O'Donoghue (1998); Harrington & O'Donoghue (1998); Hedlund (2016); Lee et al. (2009); Li et al. (2015); Madu (2010); Mitchell and Doyle (1996); Ocaña-Riola and Sánchez-Cantalejo (2005); Öğdül (2010); Peng et al. (2016); Prieto-Lara and Ocaña-Riola (2010); Romano et al. (2016); Smith and Parvin (1973); Zhao et al. (2019)	19	39%
Employment	Beynon et al. (2016); Bogdanov et al. (2008); Cromartie and Swanson (1996); Dická et al. (2019); Gajić et al. (2018); Hedlund (2016); Li et al. (2015); Ocaña-Riola and Sánchez-Cantalejo (2005); Öğdül (2010); Prieto-Lara and Ocaña-Riola (2010)	10	20%
Non-agricultural Industry Sectors	Beynon et al. (2016); Blunden et al. (1998); Bogdanov et al. (2008); Caschili et al. (2015); Cleland (1995); Gajić et al. (2018); Hedlund (2016); Leduc (1997); Li et al. (2015); Madu (2010); Mao et al. (2018); Mitchell and Doyle (1996); Nutley (1980); Öğdül (2010); Smith and Parvin (1973); Smith and Parvin (1975)	16	33%
Education	Bogdanov et al. (2008); Cleland (1995); Gajić et al. (2018); Hedlund (2016); Li et al. (2015); Madu (2010); Öğdül (2010); Romano et al. (2016)	8	16%
Personal Poverty/Wealth	Blunden et al. (1998); Bogdanov et al. (2008); Caschili et al. (2015); Cleland (1995); Gajić et al. (2018); Madu (2010); Mao et al. (2015); Mao et al. (2018); Öğdül (2010); Prieto-Lara and Ocaña-Riola (2010);	10	20%
Housing Stock	Beynon et al. (2016); Bogdanov et al. (2008); Dická et al. (2019); Madu (2010); Pászto et al. (2015); Romano et al. (2016); Shoultz et al. (2007)	7	14%
Infrastructure and Housing Quality	Cloke (1977); Harrington & O'Donoghue (1998); Li et al. (2015); Mitchell and Doyle (1996); Mountrakis et al. (2005); Ocaña-Riola and Sánchez-Cantalejo (2005); Prieto-Lara and Ocaña-Riola (2010)	7	14%
Health Services Access	Bogdanov et al. (2008); Caschili et al. (2015); Dická et al. (2019); Doogan et al. (2018); Kralj (2009); Leduc (1997); Li et al. (2015); Madu (2010); Mao et al. (2015); Mao et al. (2018); Mountrakis et al. (2005); Nutley (1980); Smith and Parvin (1973); Weinert and Boik (1995); Zhao et al. (2019)	15	31%
Quality of Life or Social/Cultural Capital	Cleland (1995); Fang and Wang (2018); Nutley (1980)	3	6%
Environmental Quality	Prieto-Lara and Ocaña-Riola (2010)	1	2%

Note: 49 studies describing a quantitative measure of rurality for which sufficient information was provided were categorized by the dimensions of rurality examined. Count of studies does not sum to 49 as most studies employ multiple dimensions.

Table 2
Construction table for studies with a quantitative rurality index or typology.

Type of Process	Specific Process	Studies	Count of Studies	Percent of Studies
Formal Statistical Processes	Factor Analysis or Principal Component Analysis	Beynon et al. (2016); Bogdanov et al. (2008); Caschili et al. (2015); Cloke (1977); Dická et al. (2019); Gajić et al. (2018); Harrington & O'Donoghue (1998); Li et al. (2015); Madu (2010); Mao et al. (2018); Mitchell and Doyle (1996); Ocaña-Riola and Sánchez-Cantalejo (2005); Öğdül (2010); Pászto et al. (2015); Prieto-Lara and Ocaña-Riola (2010); Romano et al. (2016); Shoultz et al. (2007); Smith and Parvin (1973); Smith and Parvin (1975); Zhao et al. (2019)	20	41%
	Statistical Cluster Analysis	Berry et al. (2000); Bogdanov et al. (2008); Gajić et al. (2018); Hedlund (2016); Mitchell and Doyle (1996); Nong (2015); Öğdül (2010)	7	14%
	Other (fuzzy inference and neural networks)	Blunden et al. (1998); Pászto et al. (2015); Romano et al. (2016)	3	6%
Mathematical	Linear Combinations of Multiple Variables or Multiple Factor Scores	Caschili et al. (2015); Cleland (1995); Cohen et al. (2017); Fang and Wang (2018); Hoffmann et al. (2017); Inagami et al. (2016); Kralj (2009); Lauwers et al. (2005); Leduc (1997); Lee et al. (2009); Madu (2010); Mao et al. (2015); Mao et al. (2018); Nutley (1980); Olatunde et al. (2007); Pászto et al. (2015); Peng et al. (2016); Romano et al. (2016); Smith and Parvin (1973); Smith and Parvin (1975); Waldorf (2006);; Zhao et al. (2019)	23	47%
	Other (basic classification)	Cleland (1995); Cohen et al. (2017); Cromartie and Swanson (1996); Cromartie et al. (2013); Doogan et al. (2018); Fang and Wang (2018); Hoffmann et al. (2017); Isserman (2005); Riddick and Leadley (1978); Terashima et al. (2014); Zhao et al. (2019)	11	22%
Spatial Analysis	Density and containment analysis	Blunden et al. (1998); Caschili et al. (2015); Cleland (1995); Fang and Wang (2018); Leduc (1977); Mao et al. (2015); Mao et al. (2018); Mountrakis et al. (2005); Nielsen and Johansen (2009); Terashima et al. (2014); Zhao et al. (2019)	10	20%
	Proximity analysis	Blunden et al. (1998); Cloke (1977); Cohen et al. (2017); Cromartie and Swanson (1996); Cromartie et al. (2013); Harrington & O'Donoghue (1998); Hoffmann et al. (2017); Inagami et al. (2016); Lauwers et al. (2005); Mitchell and Doyle (1996); Mountrakis et al. (2005); Nielsen and Johansen (2009); Openshaw and Coombes (1991); Smith and Parvin (1975); Waldorf (2006)	15	31%
	Network Analysis	Cromartie et al. (2013); Doogan et al. (2018); Jacquet et al. (2017); Kralj (2009); Leduc (1997); Mao et al. (2015); Mountrakis et al. (2005); Olatunde et al. (2007); Pászto et al. (2015); Zhao et al. (2019)	10	20%
	Land cover analysis	Blunden et al. (1998); Caschili et al. (2015); Fang and Wang (2018); Hoffmann et al. (2017); Lauwers et al. (2005); Leduc (1977); Mao et al. (2015); Mao et al. (2018); Nong (2015); Pászto et al. (2015); Zhao et al. (2019)	10	20%
	Other	Romano et al. (2016); Terashima et al. (2014)	2	4%

Note: Techniques such as normalizing or standardizing raw data were considered to be basic data processing techniques, not an integral part of the measure construction process, and are not included in the table above. (These techniques also are not described in many studies.) Normalization, scaling, and standardization of constructed indices are common for linear combinations, but too varied to include a detailed breakdown within the summary table.

Throughout the coding, pairs of coders met regularly and had weekly check-ins with the first author.

3. Results

3.1. History and development of "rurality" and its applications

"Rurality," conceptually and functionally in its descriptive definitions and uses, has changed over time due to increasing urbanization and technological advancements. This shift is reflected in the types of quantitative measures used by scholars and government officials to define rurality. Government definitions of rurality are actively used to establish resource allocation and policy decisions. However, they are often based solely on population metrics that maintain a simple dichotomy of urban and non-urban, also designated as rural, and do not adequately represent the inherent differences and range of differences between these spaces. Researchers have attempted to remedy these problems with varying methods of measuring rurality that vary from quantitative to qualitative and continuous to categorical. As rurality, as a concept and a measure, has changed over time so has its descriptive definitions and measurements, driven in part by technology and data.

While rurality as a concept used to delineate non-urban areas from urban areas has been studied for more than a century (Mountrakis et al., 2005), scholarly study of rurality, particularly with respect to governmental policy and implementation, began in earnest in the 1930s (Hamilton, 1930). These early definitions included new concepts outside of population metrics such as road types, leadership, community history, group meeting places and locational attitudes, and a method of constructing an index of rurality is also suggested (Hamilton, 1930). However, there was not much work within the field until a renewed interest in the 1970s when work establishing indexes of rurality that relied on a combination of population density and employment patterns began (Smith & Parvin, 1973, 1975).

The most seminal of these studies examining rurality, as a concept and a quantitative measure, was done by Paul Cloke in the 1970s. Cloke (1977) proposed a new quantitative measure of rurality that included population density, demographics, commuting patterns, migration, and distance to various sized population centers. This work provided a clear methodology for a continuous and quantitative measure of rurality and firmly established quantitative measures of rurality as central in geographic and policy studies. At the same time, use of measures of rurality also began to develop within the healthcare and medical field including consideration of census categories of urban versus rural and their relation to healthcare services in a community (Riddick and

Leadley, 1978).

Rurality research during the 1980s and 1990s was further advanced by the development and use of more modern spatial analysis and statistical techniques, including factor analysis and neural network applications (Blunden et al., 1998; Cleland, 1995; Hoggart, 1988; Nutley, 1980). The 1990s saw the publication of multiple works focused around spatial analysis techniques enabled by modern Geographic Information Systems (GIS) (Cromartie and Swanson, 1996; Harrington & O'Donoghue, 1998; Mitchell and Doyle, 1996; Weinert and Boik, 1995).

The 2000s saw the continued development of regional and national measures of rurality and an increasing emphasis placed on the context of rurality measures. For instance, the rurality of various areas in Spain, Serbia, Ontario and Nigeria was examined to more accurately capture the rural setting specific to each. In Spain, Ocaña-Riola and Sánchez-Cantalejo (2005) set out to devise a continuous, municipality-based measure of rurality specific to Spain, which was later revised by Prieto-Lara and Ocaña-Riola (2010). In Serbia, a country lacking national-level definitions of rural and urban space, Bogdanov et al. (2008) classified rural places using more than forty factors, including industry data, demographics, topography, existing infrastructure and health care, divided into eight principal components. Other national-level studies for this time period examined rurality in Norway, Turkey, and Nigeria (Hidle et al., 2006; Madu, 2010; Öğdül, 2010). In conjunction with the emphasis of the local context, rural-specific local policies and implementation were emerging, particularly in healthcare (Shoultz et al., 2007).

Post 2010 marked a time period where the need for rurality research became imperative for policy guidance. For instance, since the introduction of reform and opening-up policy in 1978, China has undergone a rapid and lasting growth in economic and technological innovations along with improved infrastructure (Chen and Feng, 2000). The Chinese government proposed a concept of "urban-rural integration" in the early 2000s, largely demonstrating its ambition on speeding up the process of urbanization. The transition from a traditional agricultural community towards a modern industrial civilization has vastly reshaped the landscape of rural and urban development across the nation (Ding, 2002). However, this substantial shift has been difficult to implement for regional rural development and has hindered the attempts of policy makers to develop feasible planning strategies (Li et al., 2015), which has caught researchers' attention in developing a more localized approach in measuring rurality in China. For example, based upon national census data for 2000 and 2010, Li et al. (2015) established an index system to evaluate the degree of rurality at a county level and furthermore evaluate the correlation of rurality and socio-economic and geographical indicators, similar to other recent studies (Peng et al., 2016; Fang et al., 2018).

At the same time, more empirical studies have been focusing on incorporating advanced techniques into studying rural classification. Using European data, researchers employed a geostatistical multi-criteria model in developing a new rurality index on the basis of identified variables that would best describe the degree of rurality (Romano et al., 2016). These spatial statistical techniques explored the indicators of rurality among social, economic and landscape-related categories. Nong (2015) identified the potential use of landscape metrics and population data for delineating urban, peri-urban and rural areas in Hanoi, Vietnam in response to the process of urbanization. By incorporating a supervised classification method with Landsat remote sensing imageries and GIS-based gradient analysis technique, the study may enable urban planners to assess and adjust in a timely manner when limited census data are available (Nong, 2015). Furthermore, in terms of other techniques, the inclusion of network analyses to determine the shortest driving distance along road networks has been discussed in previous studies (Zhao et al., 2019; Jacquet et al., 2017) but data limitations remain a challenge.

In short, rurality as a concept and a measure has developed substantially from the basic dichotomy of otherness (non-urban) to a highly complex measure that includes varied and multitudinous factors using sophisticated techniques and fine scale data. In the next sections, we examine in detail how rurality has been defined and conceptualized in the primary studies of our systematic review. Since our focus is on the application and implications of rural classification for policy and practice, we emphasize the quantitative definitions and measures of rurality, particularly in terms of the techniques, methods, and data used to create different measures of rurality.

3.2. *Conceptualizations of rurality*

At its core, rurality was conceptualized as a space that is apart from urban areas. As discussed in the previous section, rural area was originally defined as essentially non-urban. However, as society has changed and the need to better define rurality for policy and practice become more apparent, rurality has continually experienced changes in its conceptualization. There are some works that explored the more subjective perspectives of what rurality means and what it means for the people who live in rural spaces (e.g., Chigbu, 2013; Hoggart, 1988; Rowles, 1988; Manusov et al., 2010; Niskač et al., 2010). For instance, Rowles (1988) examined how aging plays a role in the rural environment, while Chigbu (2013) defined rural areas as land spaces with culturally defined identity and Manusov et al. (2010) conceptualized rurality in terms of quantified and perceived rural background. On the other hand, some studies considered rurality from an epidemiological perspective, in terms of access to amenities and employment, or as a function of resources and their interactions (Lourenço, 2012; Olsen, 2013; Virgil, 2010; Zapf, 2001). Others have examined how community bonds, religiosity, parenting philosophy, traditionalism, and individualism contribute to the decision of choosing rural life and the environment of rural life (Chalfant and Heller, 1991; Crockett et al., 2000; Gruffudd, 1994; Provorse, 1996; Willits, 1964). Many of these studies emphasized the heterogeneous and context-dependent nature of rurality, accurately pointing out that "rurality" is a fluid concept that is based on personal perceptions and mindsets as much as it is on any single universal quantitative variable. Scholars have consistently argued that rurality is a social construction that does not apply equally to all rural spaces (Halfacree, 1993; Sherval, 2009; Woods, 2009). While these studies provide a more varied, and more nuanced, examination of rurality, the majority (49 out of 65) of studies in our systematic review that examine "rurality" have done so from a quantitative perspective[3] (Fig. 1), particularly with the use of objective measures such as distance and population density.

To this point, the most common components used in creating quantitative rurality measures include population size and density, demographics (e.g., age, percent White, percent dependent, percent religious), remoteness (e.g., distance to metropolitan areas), accessibility (distance to services, access to land and naval infrastructures such as highways, railroads, and ports), various measures of agricultural production or land use (e.g., percent agriculture, percent land cover), land cover (e.g. developed area), and economic measures (e.g., unemployment rate, median family income, non-agricultural jobs, percent health

professionals, percent self-employment, education level). Table 1 provides the tabular summary of each study that uses each of these main categories of components.

Using these components, scholars and governments have created measures of rurality that can be separated into categorical or continuous measures. About 30% of the studies examined described a categorical measure of rurality, 60% described a continuous measure of rurality, and the remainder described both continuous and categorical measures. Categorical measures of rurality neatly divide each geospatial area into mutually exclusive categories or groups such as the urban-suburban-rural categorization (Cromartie and Bucholtz, 2008) and the nine categories of rural-urban continuum used in the United States (Butler, 1990; USDA ERS, 2013). Most of the early work on rurality from governmental bodies have focused on these categorical measures, mainly because they facilitate decision making for policy and practice such as providing specific resources to rural areas or providing incentives for critical jobs like farmers, healthcare providers or teachers in rural areas (Bulter, 1990).

However, as findings emerged over time, researchers began to recognize the need for more nuanced views of rurality, including more accurate and sophisticated ways of measuring rurality. For instance, "rural" areas below a certain population threshold or density, or those that may be far away from metropolitan areas may have vastly different demographics and socioeconomic structures than others. Rural areas have never been monolithic as evidenced by early qualitative work (Hamilton, 1930; Willits, 1964) and researchers and policymakers could not treat them as such. Recognizing this need, much of the research conducted after 1970 focused more on proposing and developing continuous measures of rurality or detailed typologies.

Continuous measures of rurality often used several components to create a continuous scale, often from zero to one, of rurality and they employed several different analytical techniques such as weighted and unweighted averages, as well as factor/principal component analysis. Development of continuous measures of rurality has continued be an area of growing interest and has been particularly buoyed in recent years with the expansion of advanced GIS and remote sensing technologies and data availability (Fig. 2). Typologies also frequently employed statistical clustering techniques, such as factor analysis, yet result in a classification that describes the differing character of rural areas as opposed to quantifying the degree of rurality along a single spectrum. Next we discuss how measures of rurality are constructed (summarized in Table 2) as well as the country or area of focus and unit of analysis of these measures (summarized in Table 3).

3.3. Operationalizing rurality

One of the most common methods of constructing rurality measures uses linear combinations, including weighted and unweighted linear sums and averages. About 47% of the quantitative studies examined employed a linear combination of several components or factors at some point in the construction of their measure of rurality. Some examples of studies employing weights/averages are Leduc (1977), Cleland (1995), Lauwers (2005), Waldorf (2006), Kralj (2009), and Cohen et al. (2017). For instance, Cleland (1995) constructed their rurality measure using an additive sum of the following equally weighted factors: access to metropolitan via interstate, population density, percent of all employed in retail trade, percent in professional and realty services, percent in public service, median family income, persistent poverty, number of local newspapers, percent population change, and retirement destination. Cohen et al. (2017) created their measure using unweighted averages with a minimum-maximum scaling of the individual components that include log of population size, log of population density, percent of residents in urban areas, and inverse county distance to the closest metropolitan area.

The majority of studies constructing rurality measures used factor analysis, principal component analysis (PCA), or statistical cluster analysis; 49% percent of the studies we found used one or both of these techniques (see Table 2 for more details). We discuss these together because they are similar techniques. Two of the earliest studies that employ these techniques were done by Smith and Parvin in 1973 and 1975. Smith and Parvin (1973) used PCA on population density, percent living in rural areas, total population, percent employment in agriculture, percent living on farms, average annual percent change in population, percent employment in medical and dental professions, percent employment in entertainment and recreation services, percent employment in service work. Following this work, others also used PCA incorporating additional variables such as road conditions, land and naval infrastructure, and economic measures (e.g., Cloke, 1977; Caschili et al., 2015; Dicka, Gessert and Snincak, 2019; Harrington & O'Donoghue, 1998; Madu, 2010; Montgomery, 2000; Ocana-Riola and Sanchez-Cantalejo, 2005). The most influential of these was Cloke's 1977 paper that developed an index of rurality for England and Wales using PCA. While some of the studies that employ PCA developed an index of rurality that provides a continuous value of the degree of rurality, many also used PCA or cluster analysis to develop typologies that describe rurality or rural character.

Most studies employed some form of spatial analysis such as: network analysis, proximity analysis, density analysis, and land cover analysis; ~63% of the studies examined used at least one of these techniques. Using proximity and network analysis, researchers have calculated: distances from various geographical units to urban centers, how connected areas are to other areas, travel time to urban areas or place of employment, and access to health services (Blunden et al., 1998; Cromartie et al., 2013; Hedlund, 2016; Mitchell et al., 1996; Nong, 2015; Weinert and Boik, 1995). For instance, Cromartie et al. (2013) computed travel time from each location to the edge of urban areas using network analysis then classified based on travel time to urban areas for different population sizes. Other studies (e.g. Mao et al., 2018; Mountrakis et al., 2005) used density and containment analysis to examine access and availability of services and infrastructure, while other recent studies (e.g. Hoffmann et al., 2017; Nong, 2015) have employed land cover analysis of remotely sensed imagery to evaluate the pervasiveness of developed and undeveloped land cover.

Less commonly employed techniques include complex statistical processes such as fuzzy inference (Romano et al., 2016) and neural network classification (Pászto et al., 2015), or very simple processes such visual inspection of aerial imagery (Terashima et al., 2014). About 22% of the studies used a simple mathematical process such as comparison of values to thresholds taken from the literature or breaking continuous values into quantiles, to classify continuous variables or developed rurality measures. Most of the studies examined employed multiple techniques in the construction of the measure of rurality. On average, each study employed at least two techniques, with some using only one (Berry et al., 2000) and others using up to six (Zhao et al., 2019).

3.4. Geographic location and unit of measurement

The vast majority of studies examining rurality have focused on rurality in the United States and Europe, but in the last several years, this has extended to countries in the Middle East, Africa, Southeast Asia, and China. In Table 3 and Fig. 3, we observe that about 35 percent of studies examine rurality in the United States, 28 percent in Western European countries, 8 percent in Canada and 6 percent in China. One reason why there are more studies on rurality measures in the U.S. and Europe is the

[3] This tendency for the studies captured in our review to be quantitative studies may be, in part, a reflection of a tendency towards using more location-specific terms, such as "countryside" (Gkartzios et al., 2020), as opposed to the term "rurality" for examination of rural concepts in the qualitative, socio-cultural, rural studies literature.

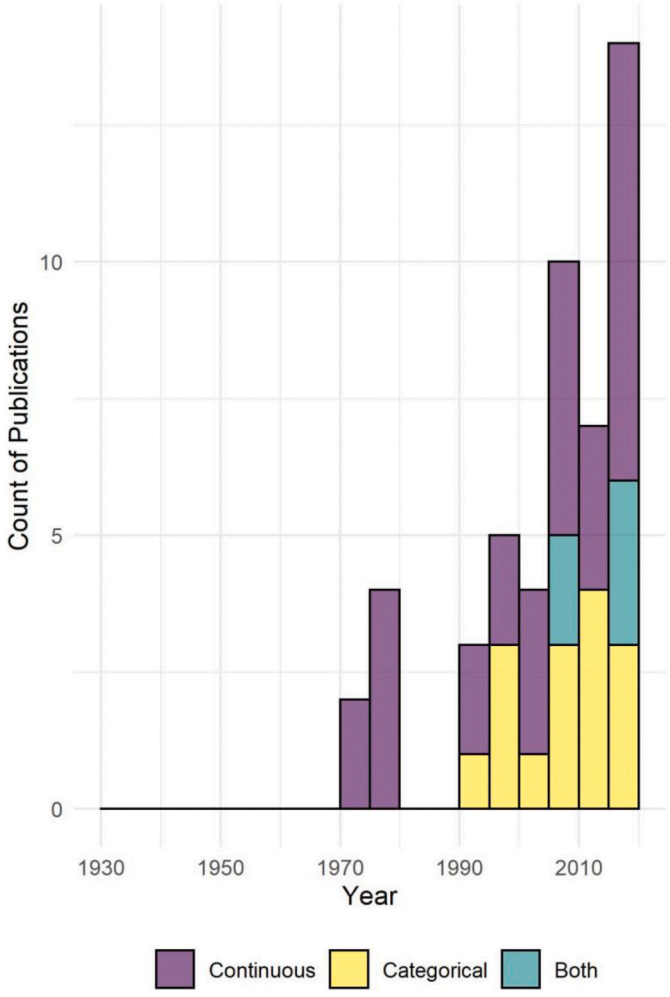


Fig. 2. Count of publications describing continuous and categorical measures of rurality over time (binned into 5-year intervals).

amount of variations in rural and urban spaces there. Moreover, up until recently, there has been limited data that can be used to study rurality in other countries and regions. However, as local census and globally available data are made more readily available more studies examining rurality in other world regions are evident (Fig. 4).

An important consideration here is the influence of early rurality measure research originating in what is now the United Kingdom. Cloke's 1977 publication has had an enormous influence on the literature with more than 400 citations in comparison to a median of 15 for the studies reviewed. While this study was not the first published study to employ principal components analysis (Smith and Parvin, 1973), it was the first to incorporate extensive demographic information and has shaped a large number of following studies. However, it is questionable to what extent rurality in the United Kingdom context is comparable with rurality in other parts of the world, as evidenced in part by the large amount of work on rurality in the United States that expanded, simplified, or customized many of the ideas from Cloke's (1977) work to U.S. contexts. As rurality research expanded into more developing areas of the world, care should be taken to not impose Western and developed country ideas and practices of rurality upon contexts in which they are not suitable. We suggest and expect there will be more studies examining rurality in the local context across the world using a combination of self-collected data, census data, and remote sensing information.

Table 3
Publications that present a rurality measure by country studied.

Country	Studies	Count of Studies	Percent of Studies
Belgium	Lauwers et al. (2005)	1	2%
Canada	Kralj (2009); Leduc (1997); Olatunde et al. (2007); Terashima et al. (2014)	4	8%
China	Fang and Wang (2018); Li et al. (2015); Peng et al. (2016)	3	6%
Czech Republic	Pászto et al. (2015)	1	2%
Denmark	Nielsen and Johansen (2009)	1	2%
India	Hoffmann et al. (2017)	1	2%
Italy	Caschili et al. (2015); Romano et al. (2016)	2	4%
New Zealand	Zhao et al. (2019)	1	2%
Nigeria	Madu (2010)	1	2%
Not Specified	Mountrakis et al. (2005)	1	2%
Serbia	Bogdanov et al. (2008); Gajić et al. (2018)	2	4%
Slovak Republic	Dická et al. (2019)	1	2%
South Korea	Lee et al. (2009)	1	2%
Spain	Ocaña-Riola and Sánchez-Cantalejo (2005); Prieto-Lara and Ocaña-Riola (2010)	2	4%
Sweden	Hedlund (2016)	1	2%
Turkey	Öğdül (2010)	1	2%
United Kingdom (England, Scotland, Wales)	Beynon et al. (2016); Blunden et al. (1998); Cloke (1977); Harrington & O'Donoghue (1998); Mitchell and Doyle (1996); Nutley (1980); Openshaw and Coombes (1991)	7	14%
United States	Berry et al. (2000); Cleland (1995); Cohen et al. (2017); Cromartie and Swanson (1996); Cromartie et al. (2013); Doogan et al. (2018); Inagami et al. (2016); Isserman (2005); Jacquet et al. (2017); Leduc (1977); Mao et al. (2015); Mao et al. (2018); Riddick and Leadley (1978); Shoultz et al. (2007); Smith and Parvin (1973); Smith and Parvin (1975); Waldorf (2006); Weinert and Boik (1995)	17	35%
Vietnam	Nong (2015)	1	2%

In terms of geographic unit of measurement, the majority of studies have used fairly large geographical units. In the U.S., the most common unit of measurement was at the county level (e.g., Smith and Parvin, 1973; Cleland, 1995; Isserman, 2005; Waldorf, 2006) with a few studies examining census tracts (Doogan et al., 2018), zip codes (Inagami et al., 2016), and smaller gridded surfaces (Cromartie et al., 2013; Mao et al., 2018). In a couple instances developed rurality measures were not for a spatial unit but were instead constructed for families (Weinert and Boik, 1995) or individuals (Mao et al., 2015) for which the impact of rurality on autonomous agency was of primary concern. In Europe, the unit of analysis has been varied from districts, to municipalities, to wards to parrishes (Harrington & O'Donoghue, 1998; Mitchell and Doyle, 1996; Ocana-Riola and Sanchez-Cantalejo, 2005). Others have examined rurality at the community settlement scale in British Columbia

(Montgomery, 2000), territorial units for statistics in Belgium (Lauwers, 2005), municipalities in Serbia (Bogdanov et al., 2008), and administrative areas in South Korea (Lee et al., 2009).

Direct comparison of the unit of analysis across regions and countries of study in terms of level of governance or area were hampered by inconsistencies in terminology and governance structures, and by a general lack of information on study unit sizes. However, when loosely organized in terms of administrative levels, it was clear that most studies developing a measure of rurality have focused on intermediate administrative levels as opposed to the highest levels of administration, which also occupied larger areas in space, and opposed to the finest levels of administration, such as the individual (Fig. 5). Much of this emphasis was practical in nature as evaluation at higher administrative levels would provide little meaningful differentiation across space while very fine scale data were in most cases unavailable and in others prohibitively difficult or time intensive to work with. In recent years unit block level census data collected by national governments and remote sensing imagery have made possible the development of rurality measures at sub-neighborhood scales (Mao et al., 2015, 2018; Zhao et al., 2019). However, as of yet these fine-scale rurality measures have only been implemented for highly developed countries (the U.S. and New Zealand) (Fig. 6), and often were not constructed for the entire country, instead focusing on one or two states (Mao et al., 2015) or territories (Zhao et al., 2019). As more advanced up-to-date and smaller scale data, such as those at the unit block level or even point level, and cell phone and remote imaging data, are made available, we expect there will be an increase in the number of studies that examine rurality at finer scales to provide more accurate information.

3.5. Uses of rurality measures

Overall, the studies indicate rurality has been conceptualized in different ways in various disciplines and interdisciplinary efforts. The studies we examined have applications within the fields of Geography, Sociology, Urban and Regional Planning, Economics, Health, Psychology, and Policy Analysis and Development. Many of the studies developed a measure of rurality for the specific purpose of exploring trends with health outcomes (e.g., Cohen et al., 2017; Doogan et al., 2018; Mao et al., 2015; Weinert and Boik, 1995; Zhao et al., 2019). Occasionally these studies employed regression analyses to validate the developed rurality measures for examining health outcomes of interest (Doogan et al., 2018), or to evaluate the associations between health outcomes and a presumed valid measure of rurality (Zhao et al., 2019). Other studies explicitly noted potential applications of the developed rurality measures to: development and analysis of urban or rural planning and resource allocation policies (e.g. Beynon et al., 2016; Bogdanov et al., 2008; Caschili et al., 2015; Cleland, 1995; Fang and Wang, 2018; Hedlund, 2016; Isserman, 2005; Openshaw and Coombes, 1991; Romano et al., 2016), issues of equity and poverty alleviation (Cleland, 1995; Li et al., 2015; Öğdül, 2010; Zhao et al., 2019), migration (Jacquet et al., 2017), education (Mao et al., 2015) and agricultural policy analysis (Mitchell and Doyle, 1996), although in most cases these links were not directly explored in the studies examined. Many of the studies focused solely on the construction and description of a rurality measure without clear discussion of how the index might be operationalized or example applications of its use in the examined study or any located follow-up published studies. It is unclear how or if these measures of rurality are being employed by actual decision-makers. For instance, the education policy literature is paying increasing attention to rural-urban distinctions, with calls for research focusing specifically on rural issues such as the AERA Open Special Topic Call for Papers on Rural Education Finance and Policy (Stoddard and Toma, 2019), however, as of yet construction of new rurality measures for education analysis purposes has not been evident. Moreover, much of the empirical literature often treats rurality simply as a control variable, even though there exists a literature that indicates how rural education is substantially different than urban education (Biddle and Azano, 2016).

The only exception to the lack of application of new measures of rurality is the Health field, where policy analysis along urban-rural lines is quite rich (Doogan et al., 2018; Mao et al., 2015; Weinert and Boik, 1995; Zhao et al., 2019). For instance, Weinert and Boik (1995) examined how various health care outcomes are correlated with measures of rurality; in particular, they found that distance to emergency care and distance to routine care were strongly associated with rurality. Similarly, Mao et al. (2015) examined how individual-based rurality measure correlates with various health outcomes for Latino immigrants. Pushing this research further along, Doogan et al. (2018) examined how their rurality measure (*Isolation*) could better explain state level health outcomes, such as infant mortality rate and smoking related mortality rate, than other measures of rurality. Even more recently, Zhao et al. (2019) developed a relative rurality measure using network analysis to examine how health outcomes vary with their multi-dimensional index in New Zealand.

4. Discussion and conclusion

By conducting a systematic review on the conceptualization, operationalization, and uses of rurality, we are able to provide a comprehensive overview of how rurality has been descriptively defined and quantitatively operationalized as well as how it has been used in policy and practice. While there have been other studies that have reviewed how rurality is defined and used in their specific field (e.g., health), to our knowledge, we are the first study to conduct a systematic review broadly focusing on the empirical nature and uses of descriptive rurality measures. However, this study is, by necessity, limited by the scope of our systematic review. In particular, our study does not provide a complete review of all the different ways rurality is conceptualized, such as subjective perspectives and socio-cultural definitions of rurality (Halfacree, 1993; Woods, 2009), as we focus on the more quantitative conceptualizations and descriptive definitions of rurality, their operationalization, and their application in research, policy, and practice. Moreover, our results are limited by the terminology employed in our search process and clearly favors studies of Western, Anglophone, cultural contexts (Gkartzios et al., 2020). We are cognizant of these limitations, and we recognize that our work does not necessarily address all aspects and conceptualizations of rurality and its characteristics. However, it does provide an overview of the rurality literature that is specifically focused on developing reproducible and quantifiable measures of rurality as well as how these measures may be used in policy and practice.

First, we described how rurality has changed over time due to urbanization as well as technological advancement and development of rurality measures. We traced the study of rurality to its origin in the academic literature, and discussed how it has been redefined and refined over time. We showed that the research on rurality conducted in the 1980s and 1990s was pushed along by the development and use of spatial analysis and statistical techniques, particularly factor analysis/PCA and GIS. Rurality research in the 21st century saw a rapid expansion of regional and national measures of rurality outside of the United Kingdom and the United States. Moreover, during the last ten years, we

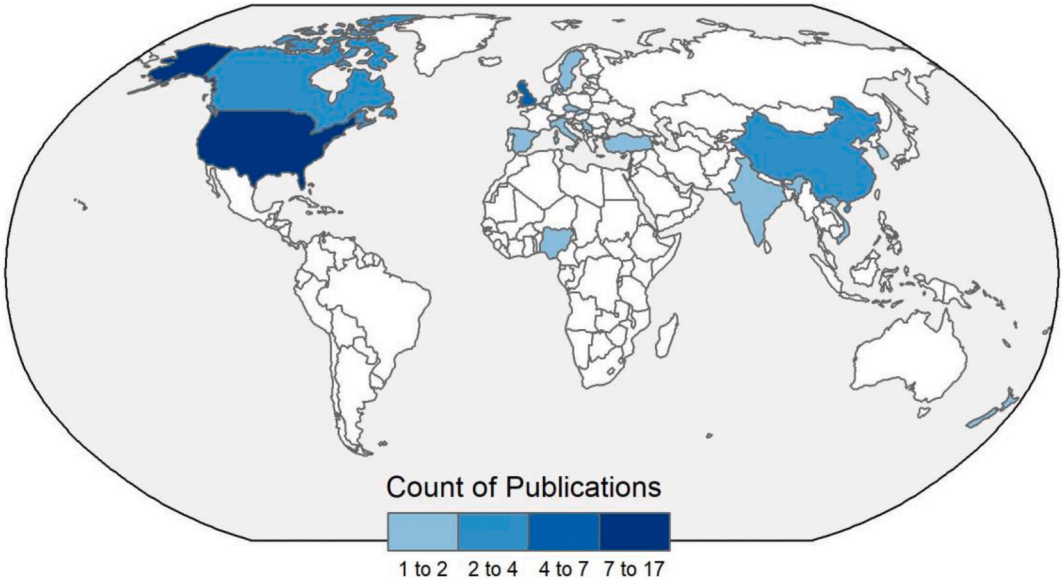


Fig. 3. Spatial distribution of publications describing a rurality measure by country studied.

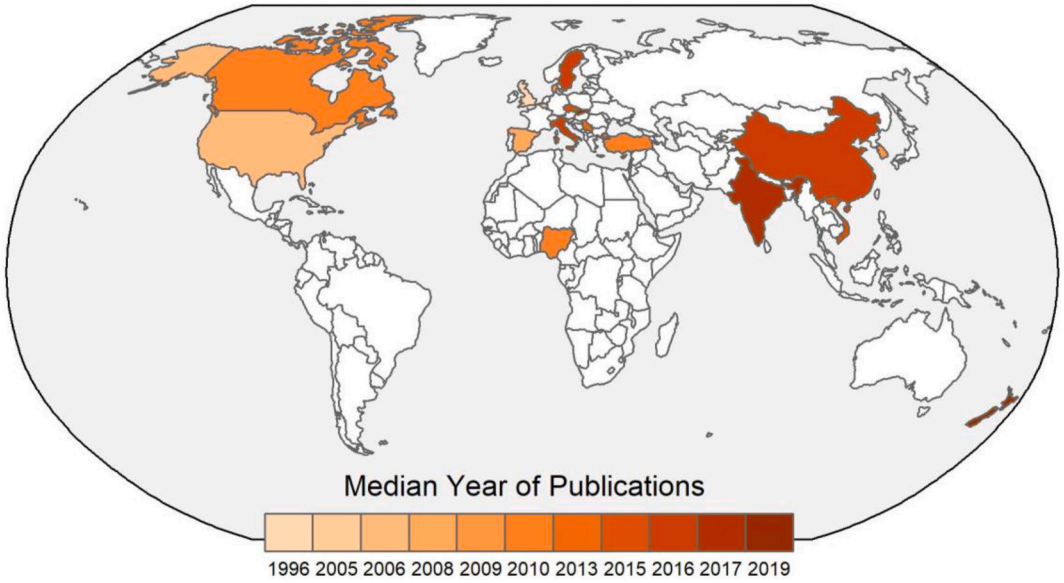


Fig. 4. Spatial distribution of rurality measure publications across countries studied by median publication year.

saw an emphasis on incorporating advanced techniques and highly detailed measurements into studying rurality and an appeal to policymakers on the use and efficacy of rurality measures.

In our results, we provided a comprehensive breakdown of the dimensions and components of rurality examined in quantitative rurality measures, as well as an overview of the techniques used in the construction of rurality measures. In addition, we provided an overview of the distribution of rurality study subjects across the globe and the scales of analysis for which rurality measures have been constructed. These results showed the strong influence of early work conducted in Western Europe on the research that followed and the development of increasingly fine-scale, complex, and data rich measures as machine learning and geographic information system technologies and data collection capabilities advanced.

In short, the study of rurality in its descriptive form has shifted from a basic dichotomy of non-urban areas to a highly complex measure that includes cutting edge data and techniques, pushing researchers and policymakers to consider how this can and should be used for policy and practice in a large number of areas ranging from urban planning to health policy to education and agricultural policy (Cohen et al., 2017; Doogan et al., 2018; Fang et al., 2018; Mao et al., 2015; Mitchell, 1996). In some ways, the foundation of how rurality is conceptualized and quantitatively measured has not changed substantively since the 1970s, but the techniques used and data that are available have made tremendous leaps. Moreover, there is more emphasis on planning, development, and policy use. At the heart of it all, in the modern era,

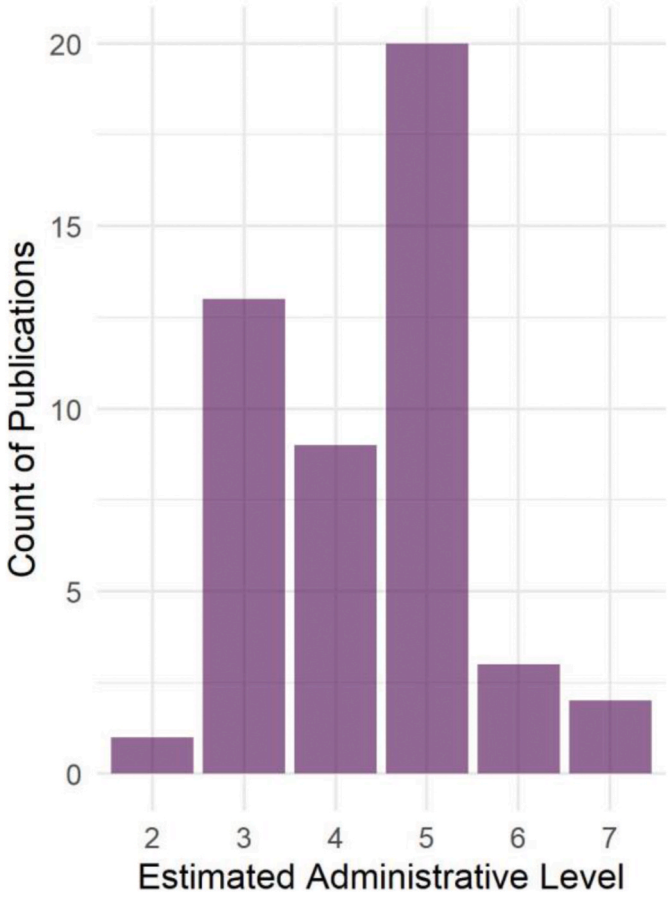


Fig. 5. Scale of rurality measure in terms of estimated administrative level. Here the primary administrative level (1) would be a National scale and the smallest administrative level (7) is the individual or family unit level. For example, Level 2 corresponds with states in the U.S. and regions in the U.K.; Level 3 corresponds to counties in the U.S., U.K., and China; Level 4 corresponds to zip codes or metropolitan areas in the U.S., districts in the U.K., and NUTS4 in Europe; Level 5 corresponds to census tracts, block-groups, or small towns in the U.S., municipalities/wards/parishes/communities in the U.K., and NUTS5 in Europe; Level 6 corresponds to small sub-neighborhood areas such as a 1 km grid.

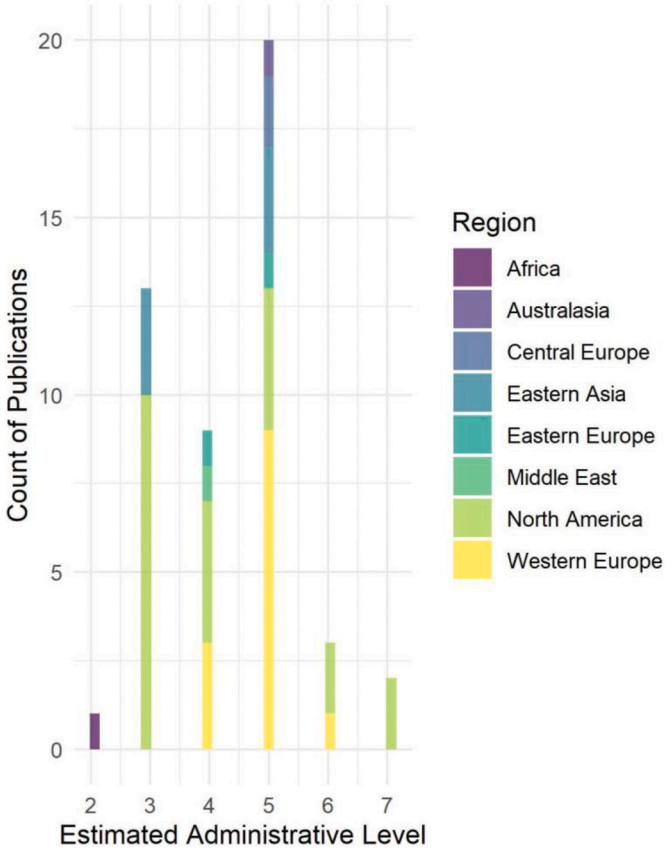


Fig. 6. Count of publications that present a measure of rurality by estimated administrative level of the measure and world region.

issues of rurality are described as issues of resource allocation, efficacy and equity.

In terms of future research, there are several areas that need further development. First, our analysis indicates while measures of rurality have become increasingly more complex and nuanced in terms of the variables considered, moving away from pure population-based measures, that some aspects of rurality remain mostly unconsidered in quantitative rurality measures and typologies. Considerations of social and cultural characteristics, environmental quality, and infrastructure quality (as opposed to quantity or proximity) have been included in only a few of the examined studies. The lack of consideration of these variables is likely due to data availability constraints, yet represents an area for growth in our understanding and characterization of rurality via integration of understandings from small-scale qualitative case studies on rurality with broad-scale quantitative rurality measures. This integration and more nuanced development would bring us closer to a more refined and inclusive conceptualization of rural space (Halfacree, 2006; National Academies of Sciences, Engineering, and Medicine, 2016). Second, more work is needed on developing measures of rurality that can be used to produce fine-scale rurality measures that better capture blurring urban-rural distinctions both within and across administrative units using widely available high-resolution spatial information for both developed and developing countries. Ideally these measures should be developed independent of demographic information for which a universal set of measures may not adequately describe all local contexts, and for which some regions will undoubtedly lack data. Relatedly, there is also a need to develop technical tools and processes that are only as complex as necessary, avoiding layering of black-box statistical techniques, and that can be used to develop these measures in consistent and reliable ways to increase practitioner and decision-maker uptake and utility. In particular, if complex statistical techniques are used, then codes, data, and replication files should be made available for other researchers and policymakers to use. If researchers are unable to make these data and codes available, then full results, including precise values of their rurality index measure should be made available online and publicly, so at the very least other researchers and policymakers may be able to leverage the work that already exists instead of having to recreate it themselves. For instance, if a researcher develops a new measure of rurality for a country or a region at the county level, then the resulting measure for all the counties under study should be freely accessible online. In short, future studies of rurality should make replication and data access more freely available to improve communication between policymakers and researchers so that these developments may

be better integrated for policy and practice.

Author contributions

K.S.N. and T.D.N. contributed equally to the study. K.S.N and T.D.N conceptualized and designed the study. K.S.N. was responsible for overall project supervision. K.S.N., N.A.B, D.G., H.C.W., J.T.W., and A.X. were responsible for data curation and analyses. T.D.N. led manuscript preparation and all authors contributed to manuscript drafting.

Declaration of competing interest

The authors declare no competing interests.

Acknowledgements

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Appendix. Tables and Figures

Table A1
Results by Database

Database	Results
ProQuest (includes dissertation and theses)	2781
Google scholar	1000
WorldCat (includes dissertation and theses)	427
JSTOR	233
Taylor & Francis	225
EBSCO	191
Web of Science	64
ERIC	29
Geobase	21
Directory of Open Access Journal (DOAJ)	9
Total	4980

Table A2
Coding Guide Schema

Study and Program Characteristics		
Variable	Description	Level of measurement
Id	ID Number assigned to study	Continuous
Leadauth	Name of lead author	Nominal
Title	Title of paper	Nominal
Yearpub	Year paper was published	Continuous
Pubtype	Type of publication (academic journal, policy report, conference paper, etc.)	Nominal
Peer review	Is the study a peer-reviewed publication?	Binary
Quantitative measure of rurality	Does the study construct or propose a quantitative measure of rurality?	Binary
How rurality is conceptualized	Description of how rurality is conceptualized	Qualitative
Geographic location	The geographic location where rurality is constructed or conceptualized	Nominal
Geographic unit of measure	The level or geographic unit of measurement used in the rural measure	Nominal
Data Source/Year	The year(s) and data source for the paper	Nominal
Type of measure of the index	The classification of the type of rural measure (categorical/classification, continuous)	Nominal
Technique/type of process	The type of process or techniques used to construct rurality (factor analysis/principal component analysis, statistical cluster analysis, linear combinations, density and containment analysis, proximity analysis, network analysis, land cover analysis, other)	Nominal
Components	Components used in the construction (population density, population, population percent/fraction in urban or rural areas, population change/migration, distance to metro/urban area, land cover (developed area, agricultural, forest or natural area), topography, network/infrastructure/service accessibility or density, age demographics, demographics, employment, industry sectors, education, personal poverty/wealth, housing stock, infrastructure and housing quality, health services access, quality of life/social/cultural capital, environmental quality)	Nominal
Policy implications	Any policy implications from the research	Qualitative
Key findings	Key findings from the study	Qualitative
Field and application	The field or area of the research	Nominal
Construction details	Additional details on the construction of rurality measure	Qualitative
Misc_note	Miscellaneous notes	Qualitative

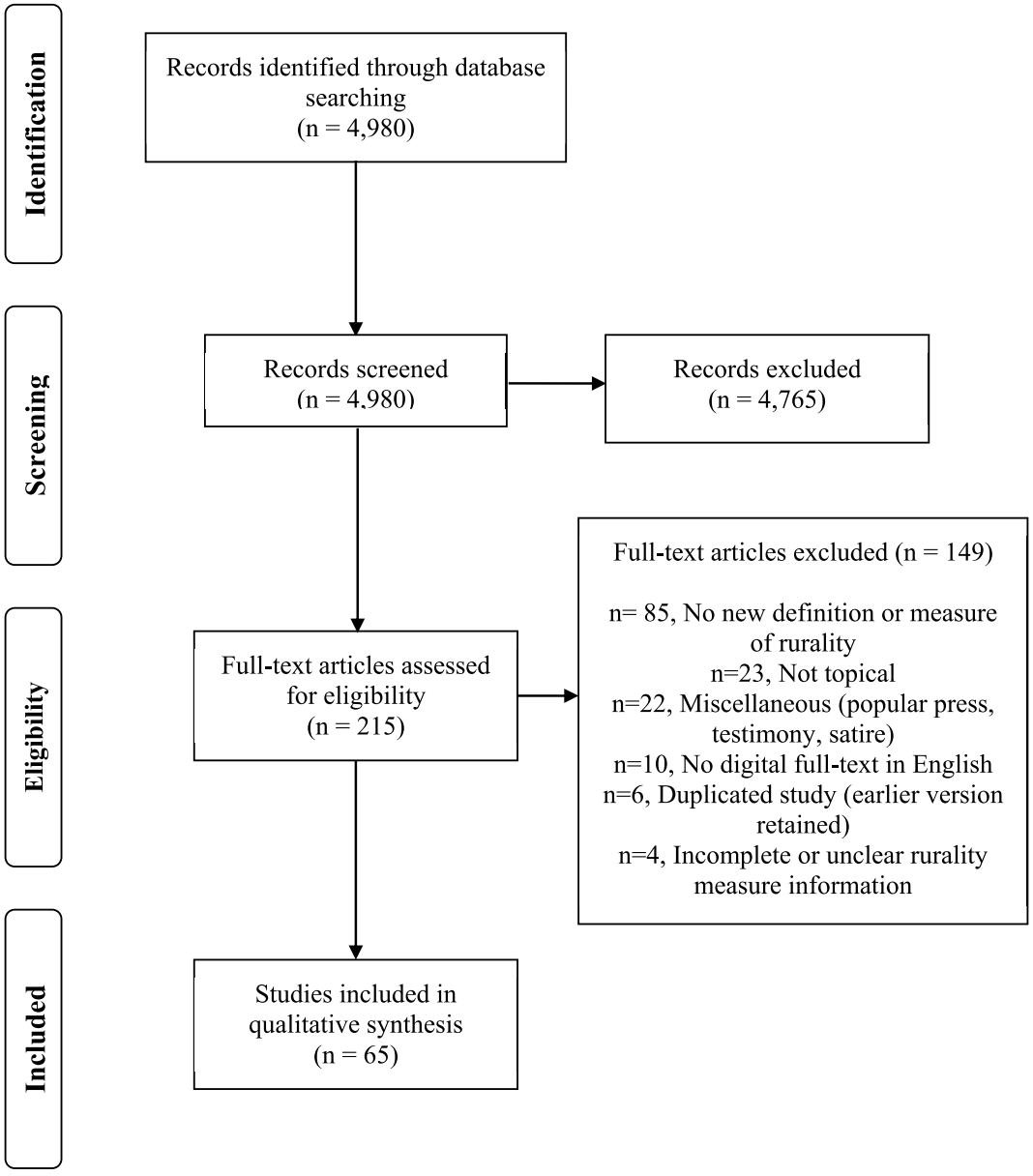


Fig. A1. Systematic review process flow chart with study counts and exclusion reasons.

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